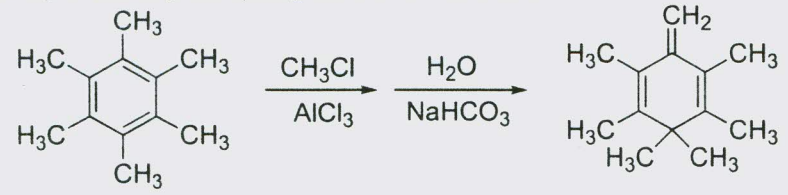
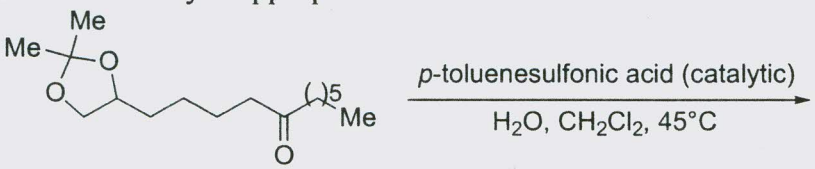
Organic Cume
4-27-2012

1 (q 12.55) Reaction of hexamethylbenzene with methyl chloride and aluminum chloride gave a salt A, which, on being treated with sodium aqueous sodium bicarbonate solution, yielded compound B. Suggest a mechanism for the conversion of hexamethylbenzene to B by correctly inferring the structure of A.

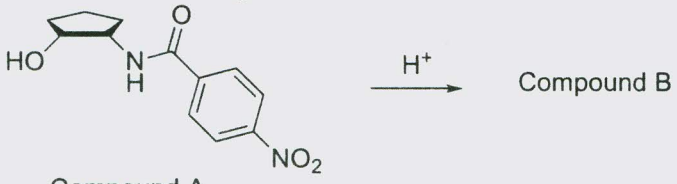


Hexamethylbenzene Compound B

2. (q 17.40) Draw the product of the following transformation and specify stereochemistry if appropriate.

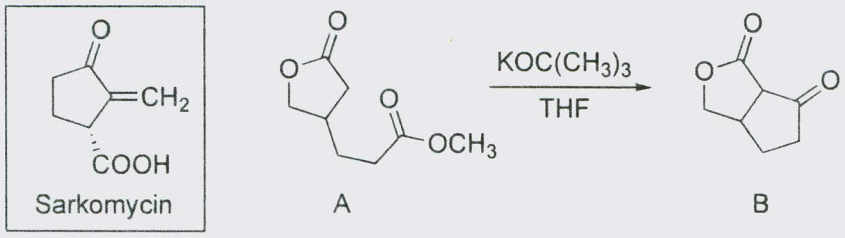


3. (q 19.46) (a) In the presence of dilute hydrochloric acid, compound A is converted to a constitutional isomer, compound B. Suggest a reasonable structure for compound B.
(b) The trans stereoisomer of compound A is stable under the reaction conditions. Why does it not rearrange?



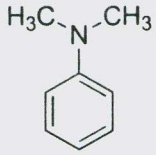
Compound A

4. (q 20.57) The α–methylene ketone sarkomycin has an inhibitory effect on certain types of tumors. A key step in the synthesis of sarkomycin is the reaction of lactone ester A with potassium *tert*-butoxide on tetrahydrofuran to give the bicyclic compound B. Write a mechanism fro this reaction.



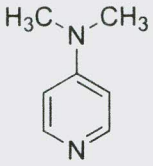
5 (q 21.53). *N,N*-Dimethylaniline and pyridine are similar in basicity, whereas 4-(*N,N*-dimethylamino)pyridine is considerably more basic then either. Apply resonance

principles to identify the more basic of the two nitrogens of 4-(*N*,*N*-dimethylamino)pyridine, suggest an explanation for its enhanced basiccity.



N,*N*-Dimethylaniline
pK_a of conjugated acid = 5.1

Pyridine
pK_a of conjugated acid = 5.3



4-(*N*,*N*-Dimethylamino)pyridine
pK_a of conjugated acid = 9.7